

Assignment 6

Logistic Distr.

$$\frac{d}{dx}(F(x; \sigma)) = \frac{d}{dx}\left(\frac{1}{1+e^{-x/\sigma}}\right)$$

$$f(x; \sigma) = \frac{d}{dx}(1+e^{-x/\sigma})^{-1}$$

$$= -(1+e^{-x/\sigma})^{-2} \cdot \frac{d}{dx}(1+e^{-x/\sigma})$$

$$= -(1+e^{-x/\sigma})^{-2} \cdot e^{-x/\sigma} \left(-\frac{1}{\sigma}\right)$$

$$= \frac{1}{\sigma} \left(\frac{e^{-x/\sigma}}{(1+e^{-x/\sigma})^2} \right)$$

Exponential Distr. $F_X(x) = \begin{cases} 0, & x < 0 \\ 1-e^{-\lambda x}, & x \geq 0 \end{cases}$

$$\frac{d}{dx}(1-e^{-\lambda x}) = \lambda e^{-\lambda x}, \quad x \geq 0$$

$$\frac{d}{dx}(0) = 0$$